

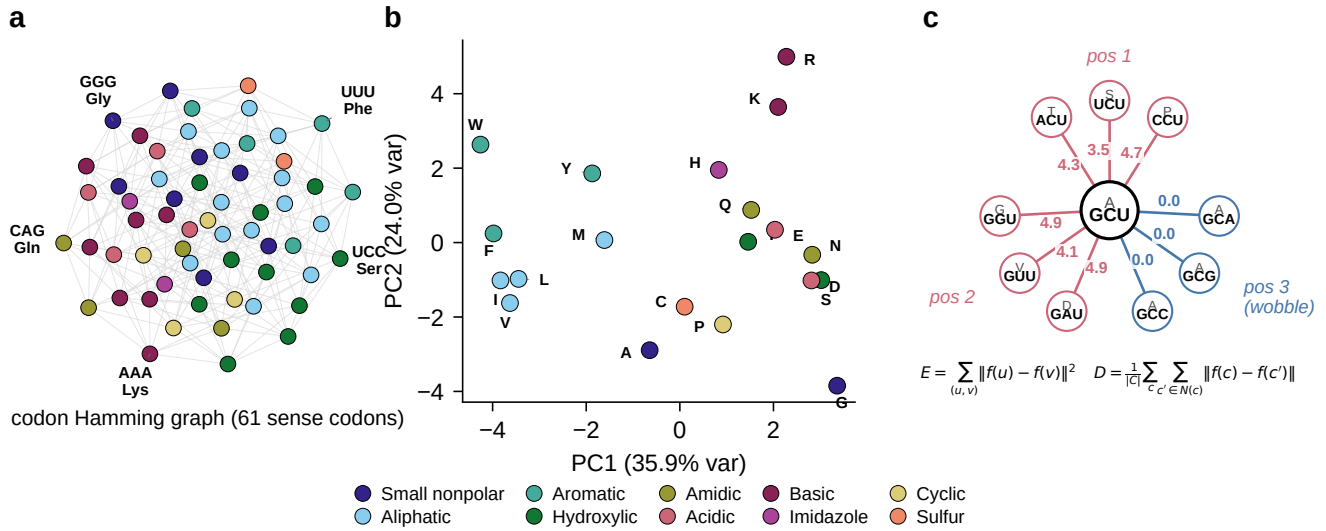


Figure 1. Framework. (a) Codon Hamming graph for $n = 3$: all 61 sense codons, with edges connecting codons that differ by a single nucleotide substitution (264 edges; average degree 8.6). Nodes are colored by amino acid chemical class (10-class RAA10 grouping, Murphy 2000); five peripheral anchor codons are labeled for orientation. (b) 20 amino acids in PC1-PC2 space of the 22-property descriptor set (after correlation-pruning and z-scoring; PC1 = 35.9 % of variance, PC2 = 24.0 %; 97.1 % total in the retained eight PCs used downstream). Hydrophobic residues (Phe, Trp, Tyr, Ile, Leu, Val, Met) lie at negative PC1; charged residues (Lys, Arg, Asp, Glu) occupy positive PC1. Class color scheme matches panel (a). (c) Hamming-1 neighborhood of the Ala codon GCU with edge-wise property distances $\|\Delta f\|$ computed in the 8-PC space: all three position-3 neighbors (GCA, GCC, GCG) are synonymous with $\|\Delta f\| = 0$ (blue edges); the six position-1 and position-2 neighbors are non-synonymous with $\|\Delta f\|$ ranging from 3.5 (UCU → Ser) to 4.9 (GGU → Gly, GAU → Asp) (red edges). Dirichlet energy E sums squared $\|\Delta f\|$ over all graph edges; noise distortion D averages $\|\Delta f\|$ over non-synonymous neighbors per codon.

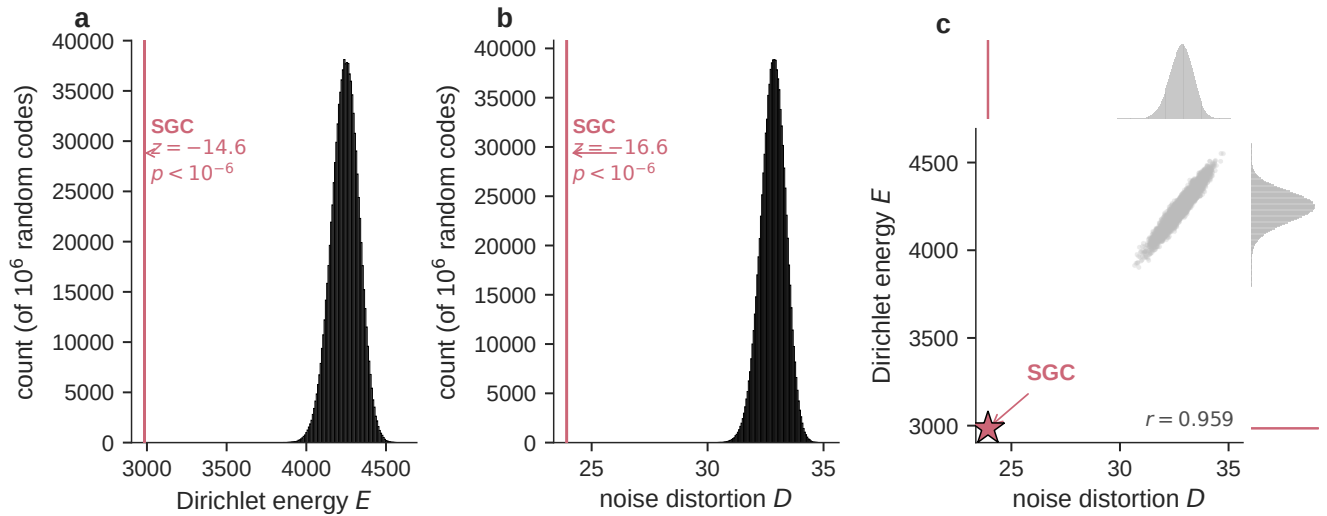


Figure 2. The Standard Genetic Code (SGC) is beyond the null under both metrics. (a) Null distribution of Dirichlet energy E across 10^6 degeneracy- preserving random codes (grey). The SGC (red line) falls at $z = -14.6$ ($p < 10^{-6}$; 0 / 10^6 random codes reach the SGC's energy). The null is generated by permuting amino acid labels across the 61 sense codons while preserving per-amino-acid codon counts, fixing stops, and pinning AUG $\rightarrow$ Met. (b) Same null-hypothesis test for noise distortion D : SGC at $z = -16.6$ ($p < 10^{-6}$; 0 / 10^6). (c) Joint scatter of (D, E) for a 10 000-point random subsample from the null distribution (grey points). The SGC (red star) lies outside and below the null cloud in both metrics simultaneously. The two metrics are strongly but imperfectly correlated across the null ($r = 0.959$), so each provides partially independent evidence; neither suggests the SGC's optimization is an artefact of the other. Marginal histograms along the top (D) and right (E) axes reproduce panels (a) and (b) for direct comparison.

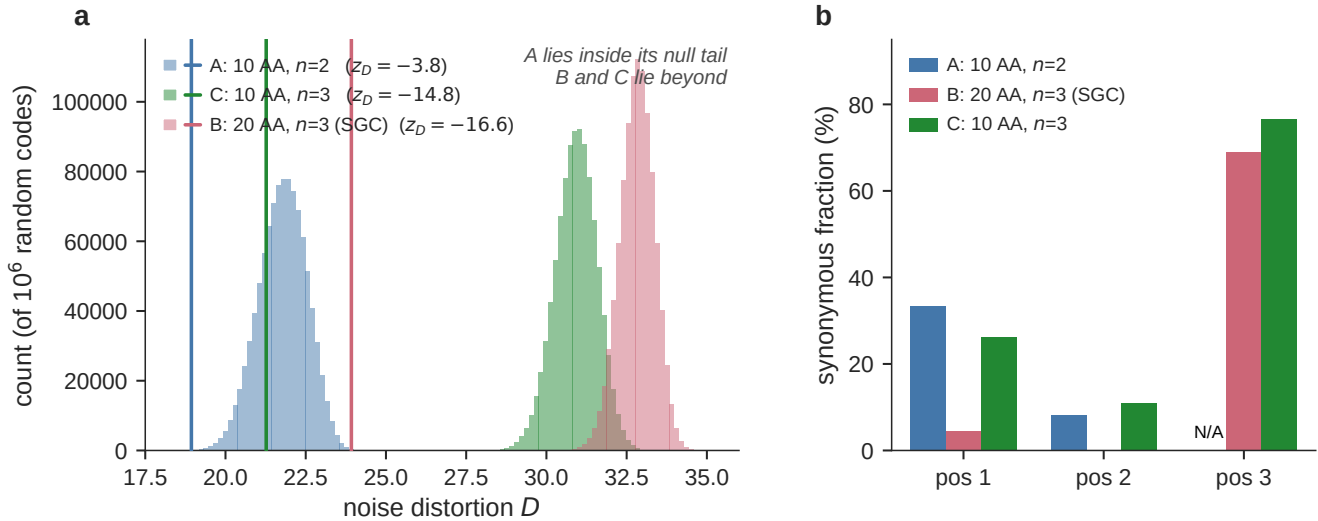
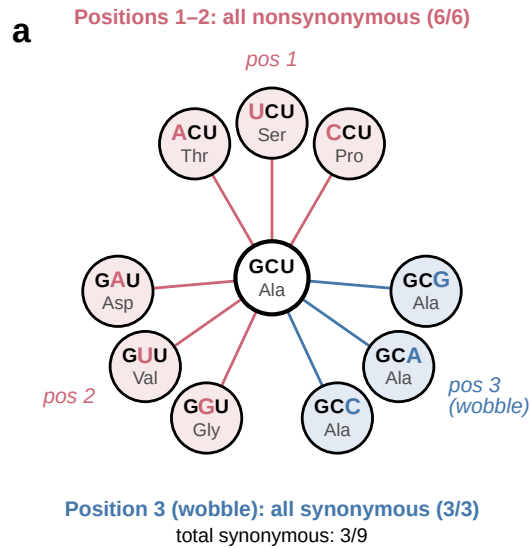


Figure 3. Triplet architecture enables deep optimization; reduced alphabet size does not. (a) Null distributions of noise distortion D across 10^6 degeneracy-preserving random codes for three conditions, on a shared x-axis: Condition A (blue; 10 amino acids, $n = 2$ doublet code obtained by majority- vote projection of the SGC onto the first two codon positions under the RAA10 reduced alphabet), Condition C (green; 10 amino acids, $n = 3$ triplet with RAA10 applied to the sense codons), and Condition B (red; 20 amino acids, $n = 3$ — the SGC itself). Vertical lines mark the observed value for the corresponding code. The doublet SGC projection (A) sits well within its null tail ($z_D = -3.8$); the triplet conditions both lie beyond their nulls ($z_D = -14.8$ for C, $z_D = -16.6$ for B). The B and C null distributions substantially overlap, indicating the architectural shift from $n = 2$ to $n = 3$ is the dominant determinant of where the observed codes sit relative to their randomized backgrounds. (b) Position-specific synonymous fraction for each condition: the fraction of single-nucleotide substitutions at each codon position that preserve the coded amino acid (or RAA class). In the triplet codes (B and C), position 3 absorbs a majority of substitutions as synonymous (68.9 % for SGC, 76.5 % for RAA10 triplet), creating an error- buffering corridor. The doublet code has no dedicated synonymy position (33.3 % at pos 1, 8.3 % at pos 2; pos 3 is absent). N/A = not applicable (condition A has no third position). Raw z-scores are reported; no inter- condition z-score differences (Δz) are computed because z-scores against different null models are not directly comparable.

Triplet: GCU (Ala) neighborhood



Doublet: GC (Ala/Gly) neighborhood

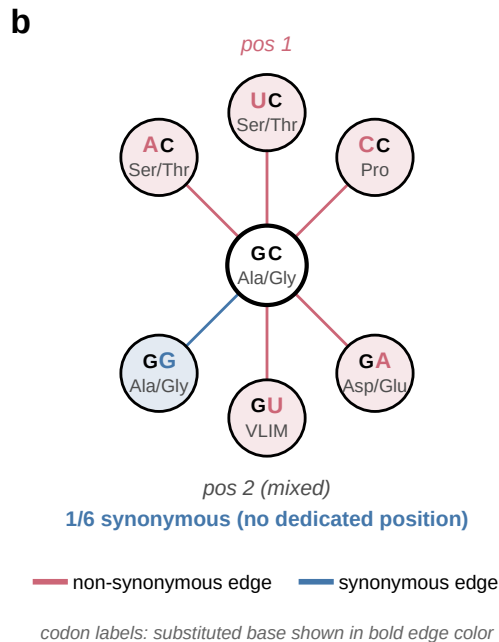


Figure 4. Wobble-position synonymy is the mechanism by which triplet architecture buffers single-nucleotide errors. (a) Hamming-1 neighborhood of GCU (Ala) in the SGC. Each neighbor differs from GCU by exactly one base (the substituted base is shown in bold in the edge color). The three position-3 neighbors (GCA, GCC, GCG) all encode Ala and are therefore synonymous with GCU (blue edges). All six position-1 and position-2 neighbors encode different amino acids (red edges). The wobble position converts all three position-3 substitutions into silent changes; positions 1 and 2 convert none, for a neighborhood synonymy rate of 3/9 (33 %). (b) Hamming-1 neighborhood of GC (Ala/Gly), the doublet-projection analogue of GCU under the RAA10 alphabet. The doublet code has six neighbors (three at each of its two positions). Only GG (which the majority-vote projection also assigns to Ala/Gly) is synonymous; the other five neighbors cross class boundaries. Synonymy rate 1/6 (17 %), with no position providing dedicated wobble tolerance. This explains the mechanism behind the architecture effect in Figure 3: the third codon position in triplet codes supplies a systematic error-absorbing channel that has no analogue in doublet codes, regardless of alphabet size.